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NUMERICAL ANALYSIS

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Numerical Analysis

5 Matrix Factorization

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5.1 LU Factorization

Example

The following identity can be checked:

$$\begin{pmatrix} -2 & -1 & -3 \\ -4 & 0 & -7 \\ 6 & 7 & 9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 2 & 1 \end{pmatrix} \begin{pmatrix} -2 & -1 & -3 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{pmatrix}$$

Let $\mathbf{A}$ be an $n \times n$ matrix. The product

$$\mathbf{A} = \mathbf{L}\mathbf{U}$$

is called LU factorization of $\mathbf{A}$ if $\mathbf{L}$ is lower triangular with all entries 1 in the main diagonal, and $\mathbf{U}$ is upper triangular.

Theorem:

Let $\mathbf{A}$ be a nonsingular square matrix. If the LU factorization of $\mathbf{A}$ exists, then its is unique.

Proof

Suppose

$$\mathbf{A} = \mathbf{L}_1\mathbf{U}_1 = \mathbf{L}_2\mathbf{U}_2$$

are two LU factorizations of the matrix $\mathbf{A}$. Since

$$\det(\mathbf{A}) = \det(\mathbf{L}_1)\det(\mathbf{U}_1) = \det(\mathbf{L}_2)\det(\mathbf{U}_2) \neq 0,$$

therefore, $\mathbf{L}_1$, $\mathbf{L}_2$, $\mathbf{U}_1$ and $\mathbf{U}_2$ are nonsingular matrices. Hence

$$\mathbf{L}_2^{-1}\mathbf{L}_1 = \mathbf{U}_2\mathbf{U}_1^{-1}.$$

It is easy to check that $\mathbf{L}_2^{-1}\mathbf{L}_1$ is lower triangular, and $\mathbf{U}_2\mathbf{U}_1^{-1}$ is upper triangular. Therefore, both are diagonal. The main diagonal of $\mathbf{L}_2^{-1}\mathbf{L}_1$ consists of only 1 entry, hence

$$\mathbf{L}_2^{-1}\mathbf{L}_1 = \mathbf{U}_2\mathbf{U}_1^{-1} = \mathbf{I},$$

which implies

$$\mathbf{L}_1 = \mathbf{L}_2 \qquad \text{and} \qquad \mathbf{U}_1 = \mathbf{U}_2.$$

Consider the definition of the Gaussian elimination. Let

$$l_{i1} = \frac{a_{i1}}{a_{11}}, \qquad i = 2, 3, \ldots, n,$$

and define the lower triangular matrix

$$\mathbf{L}_1 := \begin{pmatrix} 1 & & & & \\ -l_{21} & 1 & & & \\ -l_{31} & & 1 & & \\ \vdots & & & \ddots & \\ -l_{n1} & & & & 1 \end{pmatrix}$$

where the missing elements are all equal to 0. It is easy to check that

$$\mathbf{L}_1\mathbf{A} = \mathbf{A}^{(1)},$$

where $\mathbf{A}^{(1)}$ is the matrix obtained performing the first elimination step of the Gaussian elimination on the coefficient matrix.

It is easy to check that

$$\begin{pmatrix} 1 & & & & \\ -l_{21} & 1 & & & \\ -l_{31} & & 1 & & \\ \vdots & & & \ddots & \\ -l_{n1} & & & & 1 \end{pmatrix} \begin{pmatrix} 1 & & & & \\ l_{21} & 1 & & & \\ l_{31} & & 1 & & \\ \vdots & & & \ddots & \\ l_{n1} & & & & 1 \end{pmatrix} = \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix},$$

hence

$$\mathbf{L}_1^{-1} := \begin{pmatrix} 1 & & & & \\ l_{21} & 1 & & & \\ l_{31} & & 1 & & \\ \vdots & & & \ddots & \\ l_{n1} & & & & 1 \end{pmatrix}$$

Similarly, let

$$l_{i2} = \frac{a_{i2}^{(1)}}{a_{22}^{(1)}}, \qquad i = 3, 4, \ldots, n,$$

and define the matrix

$$\mathbf{L}_2 := \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & -l_{32} & 1 & & \\ & \vdots & & \ddots & \\ & -l_{n2} & & & 1 \end{pmatrix}$$

where all elements in the main diagonal are 1, in the second column the elements under the diagonal are $-l_{32}$, $-l_{42}$, ..., $-l_{n2}$, and all the other elements are 0. Then

$$\mathbf{A}^{(2)} = \mathbf{L}_2 \mathbf{A}^{(1)}$$

holds.

We have

$$\mathbf{L}_2^{-1} := \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & l_{32} & 1 & & \\ & \vdots & & \ddots & \\ & l_{n2} & & & 1 \end{pmatrix}$$

We define the lower triangular matrices $\mathbf{L}_3, \ldots, \mathbf{L}_{n-1}$ in a similar manner. Simple computation shows

$$\mathbf{L}_{n-1}\mathbf{L}_{n-2}\cdots\mathbf{L}_1 = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ -l_{21} & 1 & 0 & \cdots & 0 \\ -l_{31} & -l_{32} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ -l_{n1} & -l_{n2} & \cdots & -l_{n,n-1} & 1 \end{pmatrix},$$

and

$$
\begin{aligned}
\mathbf{L} &:= (\mathbf{L}_{n-1}\mathbf{L}_{n-2}\cdots\mathbf{L}_1)^{-1} \\
&= \mathbf{L}_1^{-1}\cdots\mathbf{L}_{n-2}^{-1}\mathbf{L}_{n-1}^{-1} \\
&= \begin{pmatrix} 1 & & & & \\ l_{21} & 1 & & & \\ l_{31} & 0 & 1 & & \\ \vdots & 0 & \ddots & \ddots & \\ l_{n1} & 0 & \cdots & 0 & 1 \end{pmatrix} \cdots \begin{pmatrix} 1 & & & & \\ 0 & 1 & & & \\ 0 & 0 & 1 & & \\ 0 & \vdots & \ddots & \ddots & \\ 0 & 0 & \cdots & l_{n,n-1} & 1 \end{pmatrix} \\
&= \begin{pmatrix} 1 & & & & \\ l_{21} & 1 & & & \\ l_{31} & l_{32} & 1 & & \\ \vdots & \vdots & \ddots & \ddots & \\ l_{n1} & l_{n2} & \cdots & l_{n,n-1} & 1 \end{pmatrix}.
\end{aligned} \tag{1}
$$

Let $\mathbf{U} := \mathbf{A}^{(n-1)}$, i.e., the upper triangular matrix which is the result of the Gaussian elimination. Then

$$\mathbf{U} = \mathbf{L}_{n-1} \cdots \mathbf{L}_1 \mathbf{A},$$

which gives

$$\mathbf{A} = \mathbf{L}\mathbf{U}.$$

We have proved the following result.

Theorem:

If the Gaussian elimination can be performed on a square matrix $\mathbf{A}$, then the LU factorization $\mathbf{A} = \mathbf{L}\mathbf{U}$ exists. Then $\mathbf{U}$ is the upper triangular matrix obtained by the Gaussian elimination, and $\mathbf{L}$ is defined by (1), where l_{ij} denote the factors used in the Gaussian elimination.

Example

Consider the coefficient matrix of a linear system of an earlier example in Chapter 3.

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & -1 & 2 & 4 \\ -1 & 2 & 3 & -4 \\ -2 & 1 & 4 & -2 \end{pmatrix}.$$

As we saw earlier, the Gaussian elimination can be performed on $\mathbf{A}$, and $l_{21} = 2$, $l_{31} = -1$, $l_{41} = -2$, $l_{32} = 0$, $l_{42} = -1$ and $l_{43} = 6$. If we compute the LU factorization, then we write down the Gaussian elimination so that the factors l_{ij} can be written in place of the elements which are eliminated (changed to 0):

Example cont.

$$\begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & -1 & 2 & 4 \\ -1 & 2 & 3 & -4 \\ -2 & 1 & 4 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & 3 & 6 & 8 \\ -1 & 0 & 1 & -6 \\ -2 & -3 & 0 & -6 \end{pmatrix} \sim$$

$$\begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & 3 & 6 & 8 \\ -1 & 0 & 1 & -6 \\ -2 & -1 & 6 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & 3 & 6 & 8 \\ -1 & 0 & 1 & -6 \\ -2 & -1 & 6 & 38 \end{pmatrix}$$

Now in the last matrix the elements in the main diagonal and above are the elements of the matrix $\mathbf{U}$, and the elements below the main diagonal are the entries of $\mathbf{L}$. Therefore,

$$\begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & -1 & 2 & 4 \\ -1 & 2 & 3 & -4 \\ -2 & 1 & 4 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -2 & -1 & 6 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 & -2 & -2 \\ 0 & 3 & 6 & 8 \\ 0 & 0 & 1 & -6 \\ 0 & 0 & 0 & 38 \end{pmatrix}.$$

The following results can be proved easily.

Theorem:

If all the principal minors of $\mathbf{A}$ *are nonzero, then the Gaussian elimination can be performed without row changes, and so the LU factorization* $\mathbf{A} = \mathbf{LU}$ *exists.*

Theorem:

For any invertible square matrix $\mathbf{A}$ *there exists a permutation matrix* $\mathbf{P}$ *such that the LU factorization* $\mathbf{PA} = \mathbf{LU}$ *exists.*

Suppose $\mathbf{A} = \mathbf{LU}$ is known, and consider the linear system $\mathbf{Ax} = \mathbf{b}$. Then

$$\mathbf{LUx} = \mathbf{b}.$$

We introduce the new variable $\mathbf{y} = \mathbf{Ux}$. Then the original system is equivalent to

$$\begin{aligned} \mathbf{Ly} &= \mathbf{b} \\ \mathbf{Ux} &= \mathbf{y}, \end{aligned}$$

where both systems are triangular. We solve the first equation using a forward substitution method for $\mathbf{y}$, and then the second equation using the backward substitution method for $\mathbf{x}$. It is easy to check that $n^2 + \mathcal{O}(n)$ number of multiplications/divisions are needed to solve the two triangular systems, and to compute the LU factorization, $n^3/3 + \mathcal{O}(n^2)$ number of multiplications/divisions are needed. It is especially efficient if we solve several linear system with the same coefficient matrix.

5.2 Cholesky Factorization

Let $\mathbf{A}$ be a symmetric matrix. The factorization

$$\mathbf{A} = \mathbf{L}\mathbf{L}^T$$

of the matrix $\mathbf{A}$, where $\mathbf{L}$ is a lower triangular matrix, is called the Cholesky factorization.

We note that if the Cholesky factorization exists, it is not unique. The next theorem formulates a sufficient condition for the existence of the Cholesky factorization.

Theorem:

If $\mathbf{A}$ is positive definite and symmetric, then the Cholesky factorization $\mathbf{A} = \mathbf{L}\mathbf{L}^T$ exists, the matrix $\mathbf{L}$ is real, and we can select positive elements in the main diagonal of $\mathbf{L}$.

Proof

We prove the statement using mathematical induction with respect to the dimension of the matrix $\mathbf{A}$. The statement is obvious for 1×1 matrices. Suppose the statement of the theorem holds for $(n-1) \times (n-1)$ matrices, and let $\mathbf{A}$ be an $n \times n$ matrix. We partition the matrix $\mathbf{A}$ in the following form:

$$\mathbf{A} = \begin{pmatrix} \mathbf{X} & \mathbf{y} \\ \mathbf{y}^T & a_{nn} \end{pmatrix},$$

where $\mathbf{X}$ is an $(n-1) \times (n-1)$ matrix, $\mathbf{y}$ is an $n-1$-dimensional column vector. Since $\mathbf{A}$ is positive definite, all of its principal minors are positive, hence $\mathbf{X}$ is positive definite. We are looking for the Cholesky factorization of $\mathbf{A}$ in the form

$$\mathbf{A} = \begin{pmatrix} \mathbf{X} & \mathbf{y} \\ \mathbf{y}^T & a_{nn} \end{pmatrix} = \begin{pmatrix} \tilde{\mathbf{L}} & \mathbf{0} \\ \mathbf{c}^T & d \end{pmatrix} \begin{pmatrix} \tilde{\mathbf{L}}^T & \mathbf{c} \\ \mathbf{0}^T & d \end{pmatrix}. \tag{2}$$

Here $\tilde{\mathbf{L}}$ is an $(n-1) \times (n-1)$ dimensional lower triangular matrix, $\mathbf{c}$ is an $n-1$-dimensional column vector, $d \in \mathbb{R}$.

Proof cont.

If we perform the matrix multiplication on the partitioned matrices, we get the relations

$$\mathbf{X} = \tilde{\mathbf{L}}\tilde{\mathbf{L}}^T, \qquad \tilde{\mathbf{L}}\mathbf{c} = \mathbf{y} \quad \text{and} \quad \mathbf{c}^T\mathbf{c} + d^2 = a_{nn}.$$

By the induction hypothesis the equation $\mathbf{X} = \tilde{\mathbf{L}}\tilde{\mathbf{L}}^T$ has a lower triangular solution $\tilde{\mathbf{L}} \in \mathbb{R}^{(n-1)\times(n-1)}$, where in the main diagonal we can select positive elements. This yields that $\tilde{\mathbf{L}}$ is nonsingular, so the equation $\tilde{\mathbf{L}}\mathbf{c} = \mathbf{y}$ has a unique solution $\mathbf{c}$. Let d be a (possibly complex) root of the equation $\mathbf{c}^T\mathbf{c} + d^2 = a_{nn}$. Then relation (2) holds. d can be selected to be a positive real if and only if $d^2 = a_{nn} - \mathbf{c}^T\mathbf{c} > 0$. Relation (2) implies $\det(\mathbf{A}) = \det(\tilde{\mathbf{L}})^2 d^2$. Since $\mathbf{A}$ is positive definite, it follows $\det(\mathbf{A}) > 0$. This yields that d^2 is positive, hence d can be selected to be a positive real.

Example

Find the Cholesky factorization of the matrix

$$\begin{pmatrix} 4 & -8 & 4 \\ -8 & 17 & -11 \\ 4 & -11 & 22 \end{pmatrix}.$$

We write

$$\begin{pmatrix} 4 & -8 & 4 \\ -8 & 17 & -11 \\ 4 & -11 & 22 \end{pmatrix} = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{pmatrix}$$

We consider first the equation for the first row first element:

$$4 = l_{11}^2.$$

This can be solved for l_{11}: the positive solution is

$$l_{11} = 2.$$

Example cont.

$$\begin{pmatrix} 4 & -8 & 4 \\ -8 & 17 & -11 \\ 4 & -11 & 22 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} 2 & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{pmatrix}$$

Then we consider the elements under the main diagonal of the first column:

$$\begin{aligned} -8 &= 2l_{21}, \\ 4 &= 2l_{31}. \end{aligned}$$

The solutions are $l_{21} = -4$, $l_{31} = 2$.

Example cont.

$$\begin{pmatrix} 4 & -8 & 4 \\ -8 & \boxed{17} & -11 \\ 4 & -11 & 22 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ -4 & l_{22} & 0 \\ 2 & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} 2 & -4 & 2 \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{pmatrix}$$

Now we consider the element of the main diagonal of the second column:

$$17 = (-4)^2 + l_{22}^2.$$

Its positive solution is $l_{22} = 1$.

Example cont.

$$\begin{pmatrix} 4 & -8 & 4 \\ -8 & 17 & -11 \\ 4 & \boxed{-11} & 22 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ -4 & 1 & 0 \\ 2 & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} 2 & -4 & 2 \\ 0 & 1 & l_{32} \\ 0 & 0 & l_{33} \end{pmatrix}$$

Then look at the element in the second column under the main diagonal:

$$-11 = 2(-4) + 1 \cdot l_{32}.$$

This can be solved as $l_{32} = -3$.

Example cont.

$$\begin{pmatrix} 4 & -8 & 4 \\ -8 & 17 & -11 \\ 4 & -11 & 22 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ -4 & 1 & 0 \\ 2 & -3 & l_{33} \end{pmatrix} \begin{pmatrix} 2 & -4 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & l_{33} \end{pmatrix}$$

Finally, the element in the third row and third column is

$$22 = 2^2 + (-3)^2 + l_{33}^2.$$

This gives $l_{33} = 3$.

We have then

$$\begin{pmatrix} 4 & -8 & 4 \\ -8 & 17 & -11 \\ 4 & -11 & 22 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ -4 & 1 & 0 \\ 2 & -3 & 3 \end{pmatrix} \begin{pmatrix} 2 & -4 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 3 \end{pmatrix}.$$

Algorithm: Cholesky factorization

INPUT: $\mathbf{A}$
OUTPUT: $\mathbf{L}$

$l_{11} \leftarrow \sqrt{a_{11}}$
for $i = 2, \ldots, n$ **do**
  $l_{i1} \leftarrow a_{i1}/l_{11}$
end do
for $j = 2, \ldots, n-1$ **do**
  $l_{jj} \leftarrow \sqrt{a_{jj} - \sum_{k=1}^{j-1} l_{jk}^2}$
  for $i = j+1, \ldots, n$ **do**
    $l_{ij} \leftarrow \left(a_{ij} - \sum_{k=1}^{j-1} l_{ik} l_{jk}\right) / l_{jj}$
  end do
end do
$l_{nn} \leftarrow \sqrt{a_{nn} - \sum_{k=1}^{n-1} l_{nk}^2}$
output$(l_{ij}, \quad i = 1, \ldots, n,\ j = 1, \ldots, i)$

The operation count of the Cholesky faktorization is

$$n^3/6 + \mathcal{O}(n^2)$$

number of multiplications and divisions, and

$$n^3/6 + \mathcal{O}(n^2)$$

number of additions and subtractions, and

$$n$$

number of square roots.